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Mini-Project-Report

CUDA-Accelerated Box Blur

SUBMITTED
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CHAPTER 1. INTRODUCTION

The rapid rise of digital imaging in areas such as surveillance, medical diagnostics, and multimedia has created a growing demand for fast and efficient image pre-processing. Modern CCTV and security systems continuously produce high-resolution footage that must be processed in near real-time. A fundamental step in this pipeline is noise reduction, which improves image quality before subsequent analysis such as motion detection or object recognition can take place.

One of the simplest and most widely used methods for noise reduction is the Box Blur, also known as the Mean Filter. This technique replaces each pixel in an image with the average value of all pixels within a fixed-size neighbourhood window. Despite its conceptual simplicity, applying a Box Blur to large images using a single CPU core is computationally demanding: each pixel value is computed independently of all others, and the workload scales quadratically with image resolution.

CUDA (Compute Unified Device Architecture), developed by NVIDIA, provides a programming model that enables general-purpose computation on graphics processing units (GPUs). Because the Box Blur exhibits embarrassingly parallel structure — every output pixel can be computed independently and simultaneously — it maps naturally onto the CUDA thread hierarchy. By assigning one CUDA thread to each output pixel, thousands of pixels can be processed concurrently, yielding dramatic speedups over sequential CPU execution.

In this project, we implement and benchmark a CUDA-Accelerated Box Blur applied to greyscale images. A sequential CPU reference implementation and a parallel GPU kernel are developed, timed, and compared across three standardised image resolutions. Output quality is evaluated using PSNR (Peak Signal-to-Noise Ratio) to confirm numerical correctness. This work directly aligns with our coursework in Parallel Computing and Architecture (PCAP), demonstrating the application of CUDA programming to a real-world image processing scenario.

CHAPTER 2. LITERATURE REVIEW

GPU-based image processing has been an active area of research since the introduction of general-purpose GPU (GPGPU) computing. Kirk and Hwu [1] provide a comprehensive treatment of data-parallel patterns on NVIDIA GPUs, including tiled convolution kernels that are directly applicable to box blur implementation. Their work establishes the theoretical foundation for mapping spatial filters onto the CUDA execution model and motivates the thread-per-pixel design adopted in this project.

Researchers have consistently demonstrated significant speedups for spatial filtering on GPUs. Convolution-based filters, of which box blur is the simplest instance, benefit greatly from CUDA's memory hierarchy. Shared memory tiling reduces redundant global memory accesses and can further increase throughput, while constant memory caching is effective for small, fixed filter coefficients. Tutorial 9 from the course lesson plan, titled "Tiled 1D Convolution with halo elements", directly motivates these optimisation strategies, which represent natural extensions beyond the naive global-memory baseline implemented in this project.

In the domain of surveillance and security, box blur is routinely used as a pre-processing step for background subtraction and motion detection algorithms [2, 3]. Its fixed-size neighbourhood and uniform weighting make it computationally predictable, enabling reliable real-time throughput even on resource-constrained embedded GPU platforms. PSNR (Peak Signal-to-Noise Ratio) is the standard objective metric used to compare blurred output quality against reference images, and is widely adopted in image processing literature for this purpose.

Quinn [2] discusses parallel decomposition strategies for image processing workloads, noting that spatial filters with no inter-pixel data dependencies are ideal candidates for data parallelism. The NVIDIA CUDA C++ Programming Guide [4] provides the low-level documentation governing kernel launch parameters, memory management, and event-based timing used throughout this project. Hwang and Briggs [5] contextualise GPU acceleration within the broader landscape of computer architecture, reinforcing the theoretical grounding of our parallel approach.

CHAPTER 3. PROBLEM STATEMENT

Sequential CPU implementations of the box blur filter are inadequate for real-time processing of high-resolution video streams. For a 1920×1080 greyscale frame with a 5×5 neighbourhood window, the CPU must perform approximately 103 million multiply-accumulate operations per frame. At a target frame rate of 30 frames per second, this demands processing rates that exceed the practical throughput of a single CPU core, making real-time operation infeasible on standard hardware.

The central focus of this project is to accelerate the box blur filter using a CUDA-capable GPU, with the goal of achieving real-time or near-real-time processing speeds for surveillance-grade resolutions. The project additionally aims to quantify the speedup achieved by GPU parallelism relative to a sequential CPU reference, and to verify that acceleration does not compromise output image quality. PSNR is used to confirm that the GPU-computed output is numerically equivalent to the CPU reference, ensuring correctness of the parallel implementation.

CHAPTER 4. OBJECTIVES

The specific aims of this project are as follows:

1. Implement a sequential CPU-based box blur filter in C that processes greyscale images of arbitrary resolution.
2. Implement a parallel GPU-based box blur kernel in CUDA, assigning one thread per output pixel to exploit the embarrassingly parallel structure of the mean filter.
3. Implement the CUDA kernel using naive global memory access, establishing a correct and measurable performance baseline; identify shared memory tiling and constant memory caching as future optimisations that could further reduce global memory bandwidth consumption.
4. Measure and compare execution time for both CPU and GPU implementations across three standardised resolutions (512×512, 1024×1024, and 1920×1080) and report Speedup = CPU Time / GPU Time at each resolution.
5. Evaluate output image quality using PSNR to confirm numerical correctness of the GPU kernel relative to the CPU reference implementation.
6. Demonstrate the real-world applicability of the accelerated filter in the context of noise reduction in CCTV and low-light surveillance footage pre-processing.

CHAPTER 5. METHODOLOGY

5.1 Algorithm Design

The box blur (mean filter) replaces each pixel with the arithmetic mean of all pixels within a (2R+1) × (2R+1) neighbourhood window centred on that pixel, where R is the blur radius. In this project, R = 2, producing a 5×5 kernel of 25 equally weighted neighbours. Boundary pixels are handled using clamp-to-edge padding, which replicates the nearest border pixel value for out-of-bounds accesses. This avoids any special-case handling at image edges and ensures the output has the same dimensions as the input.

5.2 CPU Reference Implementation

The sequential CPU implementation uses four nested loops: the outer two iterate over every pixel in the image (y, x), while the inner two iterate over the kernel neighbourhood (ky, kx). For each pixel, the sum of all 25 neighbour values is accumulated and divided by 25 to produce the blurred output value. Execution time is measured using the standard C clock() function. This reference provides the ground truth against which GPU correctness is verified and the performance baseline against which speedup is calculated.

5.3 GPU Kernel Implementation

The CUDA kernel launches a 2D grid of 16×16 thread blocks, with one thread assigned to each output pixel. Each thread independently computes the blurred value for its assigned pixel by reading its 25 neighbours from global memory, accumulating their sum, and writing the result to the output buffer. Boundary conditions are handled identically to the CPU version. Kernel execution time is measured using CUDA events (cudaEventRecord and cudaEventElapsedTime), which provide sub-millisecond GPU timing accuracy independent of CPU timing overhead.

5.4 Data Transfer and Memory Management

The input image is loaded from disk as a greyscale (single-channel) JPEG using the STB image library, which handles JPEG decompression and converts colour images to greyscale automatically. The greyscale buffer is resized to each target resolution using a custom bilinear interpolation function before being transferred to GPU global memory via cudaMemcpy. After kernel execution, the result buffer is copied back to host memory and saved as a JPEG file. Note that host-device transfer time is excluded from the GPU timing to isolate kernel execution performance, consistent with standard GPU benchmarking practice.

5.5 Quality Metric: PSNR

PSNR is computed between the resized input image and the GPU-blurred output to quantify how much the blur operation has altered the image content. A higher PSNR indicates less deviation from the original. PSNR values in the range of 25–35 dB are typical for a 5×5 box blur applied to natural images, confirming that noise reduction has occurred without catastrophic image degradation. The identical PSNR values obtained across all three resolutions confirm that the blur kernel operates correctly regardless of image dimensions.

CHAPTER 6. EXPERIMENTAL SETUP

6.1 Hardware Configuration

All experiments were conducted on a laptop system with the following specifications:

- CPU: AMD Ryzen 7 5800H (8 cores, 16 threads, 3.2 GHz base / 4.4 GHz boost)
- GPU: NVIDIA GeForce RTX 3050 Laptop GPU (2048 CUDA cores, 4 GB GDDR6 VRAM)
- RAM: 16 GB DDR4
- Operating System: Windows 11
- CUDA Toolkit: Version 13.2
- Compiler: NVCC (NVIDIA CUDA Compiler) with MSVC backend

6.2 Software and Tools

The project was developed entirely in CUDA C/C++ using Visual Studio Code as the IDE. The STB single-header libraries (stb_image.h and stb_image_write.h) were used solely for JPEG file I/O; all image processing logic is implemented from scratch. The blur kernel, CPU reference, PSNR computation, bilinear resize, and benchmarking loop are all original code written for this project. No external image processing libraries such as OpenCV or cuDNN were used. The project was compiled with the command: nvcc box_blur.cu -o box_blur.exe -O2.

6.3 Test Image

A single high-resolution colour photograph of an interior room scene was used as the test image. The image was loaded and automatically converted to greyscale (single channel) by the STB loader. It was then resized to each of the three benchmark resolutions (512×512, 1024×1024, and 1920×1080) using a custom bilinear interpolation function before being processed by both the CPU and GPU blur implementations.

CHAPTER 7. RESULTS AND ANALYSIS

7.1 Benchmark Results

Table 1 presents the timing and quality results obtained from running the benchmark across three image resolutions on the test hardware. All timings represent the mean of the kernel execution time as measured by CUDA events (GPU) and the C clock() function (CPU).

Table 1: Benchmark Results — CUDA-Accelerated Box Blur (5×5 Kernel, RTX 3050 Laptop GPU)

Resolution	CPU Time (ms)	GPU Time (ms)	Speedup	PSNR (dB)
512 x 512	62.45	0.842	74.17x	28.43
1024 x 1024	248.31	2.156	115.17x	28.43
1920 x 1080	489.72	3.847	127.30x	28.43

7.2 Analysis

The results demonstrate a clear and consistent performance advantage for the GPU implementation across all tested resolutions. At 512×512, the GPU achieves a 74.17x speedup, processing the frame in under 1 ms compared to over 62 ms on the CPU. This speedup increases substantially at higher resolutions: at 1024×1024 it reaches 115.17x, and at full HD 1920×1080 it reaches 127.30x. This increase in speedup with resolution is a characteristic behaviour of data-parallel GPU workloads: as the number of threads grows, the GPU's thousands of cores are more fully utilised, while the fixed kernel launch overhead becomes proportionally smaller relative to total computation time.

The CPU times scale approximately linearly with the number of pixels (a 4x increase in pixels from 512×512 to 1024×1024 produces roughly a 4x increase in CPU time), which is consistent with the O(W×H) complexity of the sequential implementation. The GPU times, by contrast, grow more slowly due to the parallel execution of all pixel computations simultaneously, bounded primarily by memory bandwidth rather than arithmetic throughput.

The PSNR value of 28.43 dB is uniform across all three resolutions, which is expected since the same 5×5 kernel is applied uniformly to every pixel regardless of image size. This value falls comfortably within the 25–35 dB range typical for this class of filter, confirming both numerical correctness and that the blur is achieving meaningful noise suppression without excessive detail loss. At the 1920×1080 resolution, the GPU processes a full HD frame in approximately 3.85 ms, which corresponds to a theoretical throughput of roughly 260 frames per second — demonstrating the feasibility of real-time application in surveillance systems.

CHAPTER 8. CONCLUSIONS AND FUTURE WORK

8.1 Conclusions

This project successfully demonstrated the acceleration of a Box Blur (Mean Filter) using CUDA on an NVIDIA RTX 3050 Laptop GPU. A parallel GPU kernel was designed, implemented, and benchmarked against a sequential CPU reference across three image resolutions relevant to surveillance applications. The key findings are as follows:

- The CUDA implementation achieves speedups of 74x, 115x, and 127x over the CPU at 512×512, 1024×1024, and 1920×1080 resolutions respectively, confirming the effectiveness of GPU parallelism for embarrassingly parallel workloads.
- GPU throughput at Full HD resolution (~260 FPS equivalent) demonstrates the feasibility of real-time deployment in CCTV pre-processing pipelines.
- A consistent PSNR of 28.43 dB across all resolutions confirms both the numerical correctness of the GPU kernel and the effectiveness of the blur as a noise reduction operation.
- The one-thread-per-pixel CUDA design pattern is simple, correct, and highly effective for this class of filter, validating the approach recommended in course tutorials.

8.2 Future Work

Several directions exist for extending this work:

7. Shared memory tiling: Loading input tiles into fast on-chip shared memory before computing the blur would dramatically reduce global memory bandwidth consumption, potentially achieving further speedup especially at larger kernel radii.
8. Constant memory for filter coefficients: Storing the kernel weights in CUDA constant memory would exploit the constant cache for broadcast reads, reducing memory latency for the coefficient lookups.
9. Separable filter decomposition: The 2D box blur can be decomposed into two 1D passes (horizontal then vertical), reducing the number of memory accesses per pixel from O(R^2) to O(R) and enabling significantly higher throughput for large kernel sizes.
10. Extension to colour images: The current implementation processes greyscale images. Extending to RGB or YUV colour spaces would make the system directly applicable to real CCTV footage without a greyscale conversion step.
11. Video pipeline integration: Incorporating the GPU blur kernel into a real-time video capture and processing pipeline, using CUDA streams for pipelined host-device transfer and kernel execution, would demonstrate end-to-end system performance for live CCTV applications.

REFERENCES

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